

**GYROFLOX
TABLETS**
(Enrofloxacin Tablets 50 mg)

COMPOSITION:

Each uncoated tablet contains:

Enrofloxacin B.P. (Vet.): 50mg
Excipients : q.s.

Not for Human Use For Animal Treatment Only "Wash hands after use"
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DESCRIPTION:

Light yellow coloured circular tablets with a break line on one side.

PHARMACODYNAMICS

Enrofloxacin is a synthetic fluoroquinolone antibiotic that exerts its activity by inhibiting topoisomerase II, an enzyme involved in the mechanism of bacterial replication. Enrofloxacin exerts bactericidal activity concentration-dependent with similar values of minimal inhibit concentration and minimal bactericide concentrations. It also possesses activity against bacteria in the stationary phase by an alteration of the permeability of the outer membrane phospholipid cell wall.

In general, enrofloxacin exhibits good activity against most gram-negative bacteria, especially those of the enterobacteriaceae. *Escherichia coli*, *Klebsiella spp.*, *Proteus spp.*, and *Enterobacter spp.* are generally susceptible.

Pseudomonas aeruginosa is variably susceptible and, when it is susceptible, usually has a higher MIC than other susceptible organisms. *Staphylococcus aureus* and *Staphylococcus intermedius* usually are susceptible. *Streptococci*, *enterococci*, anaerobic bacteria can generally be considered resistant. Induction of resistance against quinolones can develop by mutations in the gyrase gene of bacteria and by changes in cell permeability towards quinolones.

PHARMACOKINETICS

Enrofloxacin is approximately 100% bioavailable after oral administration. It is unaffected by food. Enrofloxacin is rapidly metabolized to form an active compound, ciprofloxacin.

After a dose of 5 mg/kg body weight, maximum plasma levels of approximately 0.9 µg/mL in dogs and approximately 1.6 µg/mL in cats are reached after 1 to 2 hours.

Enrofloxacin is primarily excreted via the kidneys. A major portion of the parent drug and its metabolites is recovered in urine.

Enrofloxacin is widely distributed in the body. The tissue concentrations are often higher than the serum concentrations. Enrofloxacin crosses the blood-brain barrier. The degree of protein binding in serum is 14% in dogs and 8% in cats. The half-life lies between 2 to 7 and 4 to 7 hours for dogs and cats, respectively.

Approximately 25% of the dose of enrofloxacin is excreted in the urine and 75% via the faeces. Approximately 60% (dogs) or 15% (cats) of the dose is excreted as unchanged enrofloxacin in the urine and the remainder as metabolites, amongst others ciprofloxacin. The total clearance is approximately 9 mL/minute/kg bodyweight.

TARGET SPECIES:

Cats & Dogs

INDICATION:

Treatment of bacterial infections of the alimentary, respiratory and urogenital tracts, skin, wound infections and otitis externa.

DOSAGE:

The dose rate of Enrofloxacin is 5mg/kg bodyweight. One 50mg tablet per 10kg bodyweight is recommended.

Treatment is generally administered over 5-10 consecutive days. The recommended dosage should not be exceeded. Treatment should be re-evaluated if no improvement is seen. It is commonly advised to re-evaluate the treatment if no clinical improvement is observed within 3 days.

CONTRAINDICATION:

- Do not use in young or growing dogs (dogs aged less than 12 months (small breed) or less than 18 months (large breed)) as the product may cause epiphyseal cartilage alterations in growing puppies.
- Do not use in young, growing cats, because of the possibility of the development of cartilage lesions (cats aged less than 8 weeks or weighing less than 1kg). Do not use in cats or dogs that have seizure disorders, since enrofloxacin may cause CNS stimulation.
- Do not use in cats or dogs with known hypersensitivity to fluoroquinolones or to any of the excipients of the product.
- Do not use in case of resistance to quinolones, as there exists almost complete cross resistance to other quinolones and complete cross resistance to other fluoroquinolones.
- Do not use with Tetracyclines, Phenicolis or Macrolides because of potential antagonistic effects.
- Do not use for prophylaxis.

WARNINGS AND PRECAUTIONSSpecial warnings for each target species

None

Special precautions for use in animals

Official and local antimicrobial policies should be taken into account when the product is used.

Fluoroquinolones should be reserved for the treatment of clinical conditions which have responded poorly, or are expected to respond poorly, to other classes of antimicrobials.

Whenever possible, fluoroquinolones should only be used based on susceptibility testing.

Use of the product deviating from the instructions may increase the prevalence of bacteria resistant to fluoroquinolones and may decrease the effectiveness of treatment with other quinolones due to the potential for cross resistance.

Use the product with caution in cats or dogs with severe renal or hepatic impairment.

Pyoderma is mostly secondary to an underlying disease. It is advisable to determine the underlying cause and to treat the animal accordingly.

Special precautions to be taken by the person administering the veterinary medicinal product to animals

People with known hypersensitivity to (fluoro)quinolones should avoid any contact with the product.

In case of accidental ingestion, seek medical advice immediately and show the package leaflet to the physician.

Wash hands after handling the product.

In case of contact with the eyes, rinse immediately with plenty of water.

INTERACTIONS WITH OTHER MEDICAMENTS

- Concurrent use of Flunixin should be under careful veterinary monitoring, as the

interactions between these drugs may lead to adverse events related to delayed elimination.

- Concomitant administration of theophylline requires careful monitoring as serum levels of theophylline may increase.
- Concurrent use of magnesium or Aluminium containing substances (such as antacids or sucralfate) may reduce absorption of enrofloxacin. These drugs should be administered two hours apart.
- Do not administer simultaneously with Tetracyclines, Phenicolis or Macrolides because of potential antagonistic effects.
- Do not administer simultaneously with non-steroidal anti-inflammatory drugs, convulsions can occur.

PREGNANCY AND LACTATION:

- Studies in laboratory animals (rat, chinchilla) have not produced any evidence of a teratogenic, foetotoxic, maternotoxic effect. Use only according to the benefit/risk assessment by the responsible veterinarian.
- Enrofloxacin passes into the maternal milk, the use is not recommended during lactation.

ADVERSE DRUG REACTIONS:

- In very rare cases hypersensitivity reactions (allergic skin reactions, anaphylaxis) can occur. In these cases, administration should be discontinued and a symptomatic treatment given.

Cats:

- In rare cases, mild gastrointestinal symptoms (diarrhoea and vomiting) may occur after administration of the product.

Dogs:

- In very rare cases, possible joint cartilage alterations in growing puppies (see A11: contraindications) can be seen. In rare cases vomiting and anorexia are observed.
- The frequency of adverse reactions is defined using the following convention:
 - ✓ very common (more than 1 in 10 animals treated displaying adverse reaction(s))
 - ✓ common (more than 1 but less than 10 animals in 100 animals treated)
 - ✓ uncommon (more than 1 but less than 10 animals in 1,000 animals treated)
 - ✓ rare (more than 1 but less than 10 animals in 10,000 animals treated)
 - ✓ very rare (less than 1 animal in 10,000 animals treated, including isolated reports).

OVERDOSAGE:

- Overdosing can cause vomiting and nervous signs (muscle tremor, incoordination and convulsions) which may require treatment discontinuation.
- In the absence of any known antidote, apply drug elimination methods and symptomatic treatment.
- If necessary, administration of Aluminium- or Magnesium-containing antacids or activated carbon can be used to reduce absorption of enrofloxacin.
- According to literature, signs of overdosage with Enrofloxacin in dogs such as inappetence and gastrointestinal disturbance were observed at approximately 10 times the recommended dose when administered for two weeks. No signs of intolerance were observed in dogs administered 5 times the recommended dose for a month.
- Do not exceed the recommended dose. In case of overdose, vomiting, diarrhoea and CNS/behavioural changes may occur, this will stop when correct dosing is resumed.
- Retinal toxic effects, including blindness, may occur if the recommended dosage is

exceeded in cats.

- In laboratory studies, ocular adverse effects in cats have been observed from 20 mg/kg. The toxic effects on the retina caused by overdosing may be such that they lead to irreversible blindness in the cat.

STORAGE: Store below 25°C (air conditioning)

PRESENTATION:

10 Amber ALU/PVC Blisters packed into an outer carton with pack insert

Manufactured by:

INDIAN IMMUNOLOGICALS LIMITED

(A wholly owned subsidiary of NDDB)

Survey No. 281-284 and 321,

Biotech Park Phase – III,

Karkapatla Village, Markook Mandal,

Siddipet District, Telangana, India, Pin Code- 502281.

Product Registration Holder:

KM Vet Pharm Sdn Bhd,

No.1, JLN Industri Batu Caves 1/1,

TMN Perindustrian Batu Caves 68100,

Batu Caves, Malaysia.

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